

Decoding Customer Decisions: A Comprehensive Analysis of Revenue, Behavior and Churn in Telecom Services

Unlocking Customer Value in Telecom Through Analytics

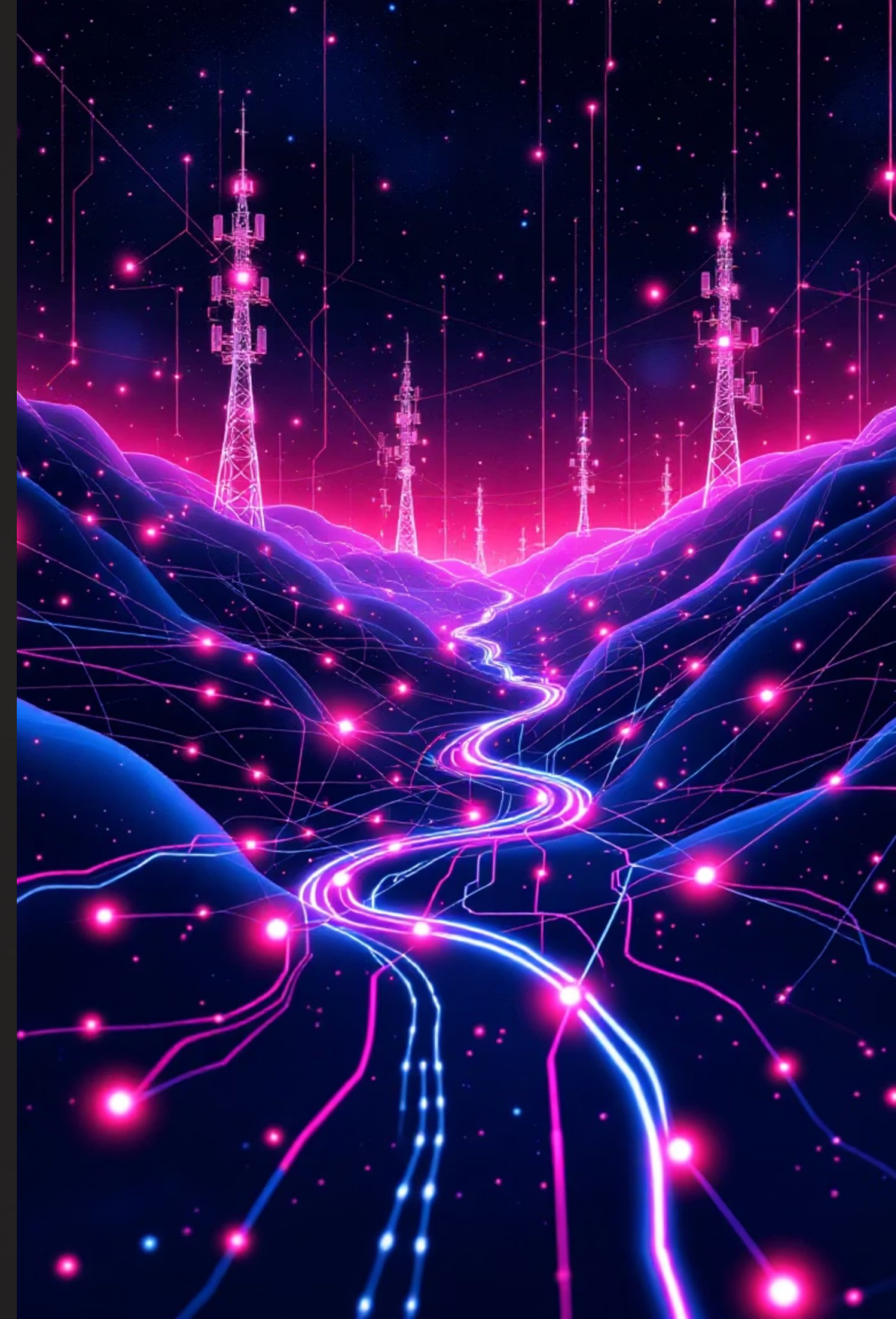
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Tool Used: Tableau

Dataset: Telecom Customer Churn

Source: Maven Analytics

Link <https://mavenanalytics.io/data-playground/telecom-customer-churn>



Our Journey Today

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The Current Reality

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Why This Analysis Matters

Telecom customers exhibit complex patterns across spending, tenure, data usage, refunds and churn status. Understanding these patterns is not just analytical, it's a business necessity.

My Objectives

- Identify high-value customers
- Understand churn behavior
- Optimize pricing & service plans
- Strengthen customer relationships



The Current Reality

Churned - 27%, Joined - 6% and Stayed - 67%

Revenue Concentration

Total revenue is strong but concentrated, with a few customers contributing disproportionately.

Service Quality

Refunds are negligible, indicating high service quality and stable operations.

Tenure Impact

Tenure has a predictable upward effect on monthly spending.

Customer Segments

Customer status reveals strong differences in lifetime value.

This is the current state of the business which seems stable but uneven



The Three-Lens Analytical Approach



Univariate

- Focuses on one variable at a time
- Shows baseline patterns and concentration of value.
- Establishes “where we are now” before exploring deeper relationships



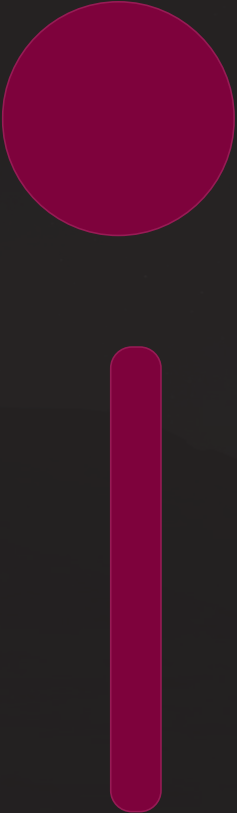
The Three-Lens Analytical Approach



Bivariate

- Examines relationships between two variables.
 - Reveals correlations and customer behavior patterns
 - Highlights segment differences and value drivers
- 

Key Identifications



Mid-Tenure Customers Stuck at \$60 of Average Monthly Spend (Visual: Average of Monthly Charge by Tenure Line Chart)

Deepen customer relationships through personalized experiences
Indicates customers are **not discovering additional value** after their first year revenue.

- This risk directly supports **Recommendation : Implement a Tenure-Based Loyalty & Upgrade Program.**
- CRM automation, reward-tier setup, promotional content and analytics tracking this would be \$50,000 as a Initial cost.

Key Identifications

Newly Joined customers and lowest charges

Newly joined customers show the lowest average total charges, signaling poor early monetization. This suggests issues in early value discovery, onboarding clarity, and product adoption.

- Supports Recommendation : Implement a Structured New-Customer Onboarding Journey to build value early and raise first-year revenue.
- Onboarding content, automation tools and customer education videos can be done under \$40,000.



The Three-Lens Analytical Approach



Multivariate

- Analyzes three or more variables simultaneously
- Reveals hidden clusters and predictive drivers
- Explains the underlying system that influences revenue and churn

Key Identifications

Extremely Slow Increase in Extra Data Consumption

(Visual: Average of Extra Data Charges + Long Distance Charges with Trend Line)

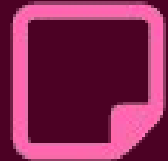
The **p-value of 0.0009** confirms the trend is statistically significant but shows a **weak growth** in extra data usage.

Slow growth in data consumption = **lost revenue**, because data usage is a major driver of premium upgrades.

- Supports **Recommendation : Expand High-Value Premium Bundles.** targeting customers by internet type and usage baseline to stimulate upgrade behavior.
- Bundle redesign, pricing evaluation, marketing campaigns which can be around \$75,000.

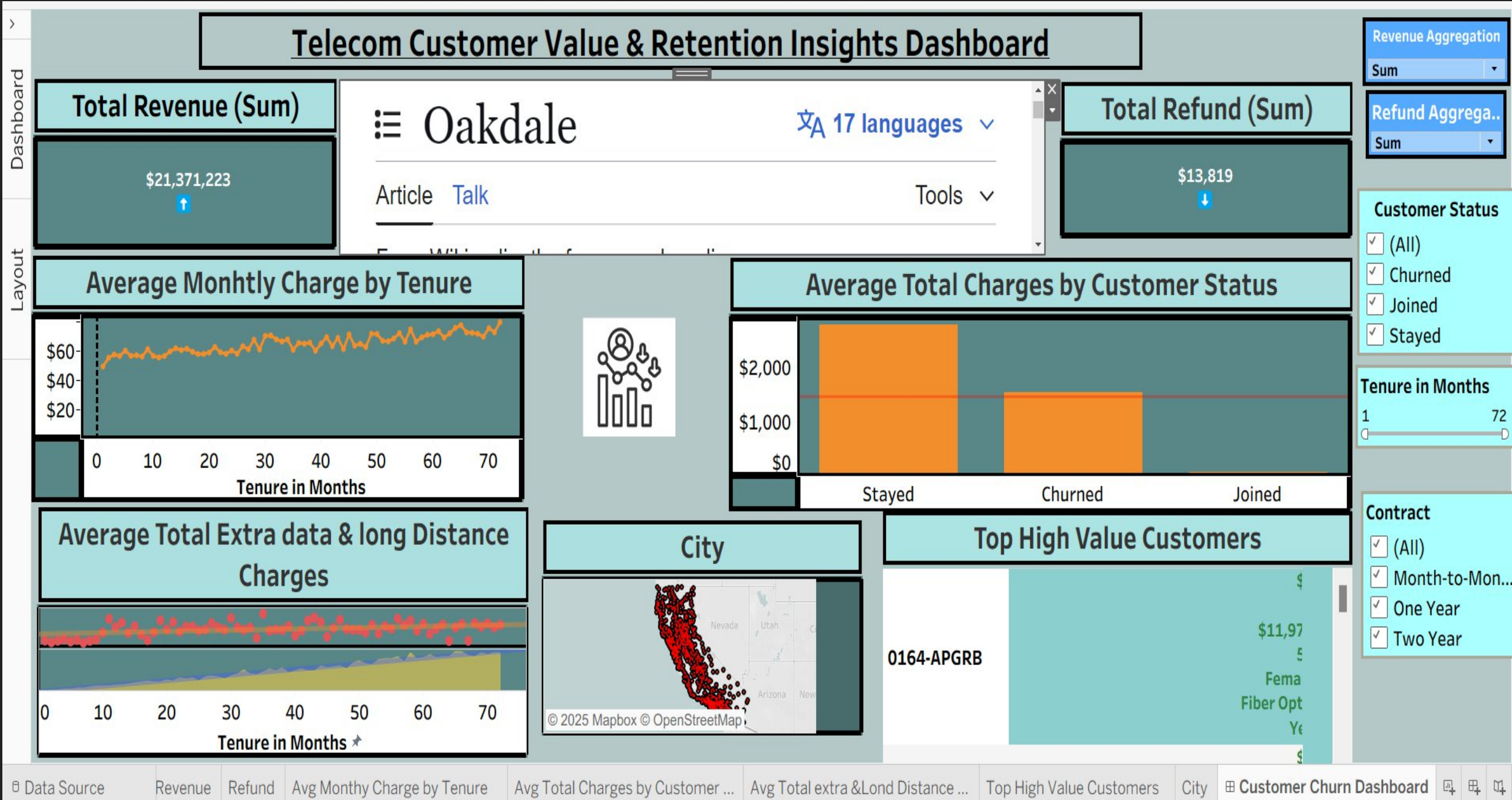
The One Truth

Customer tenure is the strongest driver of revenue, usage loyalty and long-term business value.



The company's biggest opportunity is not acquisition, it is maximizing the value of existing customers through strategic retention and upselling.







The Hidden Opportunity

Tenure as Predictor

Tenure emerges as the strongest, most reliable predictor of spending and loyalty.

Early Warning Signals

High-value churners display early warning signals in spending patterns.

Exponential Growth

Long-distance charges and data usage expand exponentially with tenure.

Premium Profile

Top customers share similar profiles: high usage + long tenure + premium plans.

Retaining and nurturing mid-tenure customers can unlock massive revenue.

Strategic Recommendations

Customer Relationships,
Revenue Streams and
Customer Segments.



Tenure-Based Loyalty Program

Implement upgrade programs to boost revenue, reduce mid-tenure churn, and convert customers into long-term revenue sources.

Impact: Generates about \$1M+ in long-term value

Value Proposition, Revenue Streams and Cost Structure.



Launch a Structured New-Customer Onboarding Journey

Target newly joined customers with personalized outreach, billing reviews, and some tailored offers.

Impact: increasing early-stage customer engagement can generate an additional \$75-\$120 in first-year revenue per new customer

Channels, Customer Relationships and Revenue Streams.



Premium Bundle Expansion

Create enhanced bundles (unlimited data + fiber + long-distance) matching evolving usage patterns.

Impact: Raises lifetime value and improves satisfaction



Addressing the Objections

Before we move forward, let's address the critical questions that shape our decision-making process and validate our strategic direction.

The Revealing outcome



These insights directly impact revenue, customer retention and future growth, the metrics that define our success.



Validated using 7,044 customers with statistically significant trends and cross-lens validation.



Implement retention programs, targeted outreach and bundle expansion in the next 3–6 months.



Now is the moment to act

Invest in customer-centric strategies that strengthen retention and unlock long-term growth.



Let's commit to action

We have the insights. We have the opportunity. Now we need decisive action.

Customer-First Initiatives

Secure long-term profitability

Industry Leadership

Position ourselves ahead of competitors

Unified Direction

Move forward as one organization

THANK YOU

